

APPROKSIMERE DERIVERTE

$$u(x+h) = \sum_{n=0}^{\infty} \frac{u^{(n)}(x)}{n!} \cdot h^n = u(x) + u'(x)h + \frac{u''(x)h^2}{2} + \frac{u'''(x)h^3}{6} + O(h^4)$$

$$u(x-h) = \sum_{n=0}^{\infty} \frac{u^{(n)}(x)}{n!} \cdot (-h)^n = u(x) - u'(x)h + \frac{u''(x)h^2}{2} - \frac{u'''(x)h^3}{6} + O(h^4)$$

$$u'(x) = \frac{u(x+h) - u(x)}{h} + O(h^2) = \frac{u(x+h) - u(x)}{h} + O(h) \quad \text{FORWARD DIFFERENCE}$$

$$u'(x) = \frac{u(x) - u(x-h)}{h} + O(h^2) = \frac{u(x) - u(x-h)}{h} + O(h) \quad \text{BACKWARD DIFFERENCE}$$

APPROKSIMERE DOBBELTDERIVERT

$$\begin{aligned} u(x+h) + u(x-h) &= 2u(x) + h^2 u''(x) + \frac{2h^4 u^{(4)}(x)}{4!} \\ &= 2u(x) + h^2 u''(x) + O(h^4) \end{aligned}$$

$$u''(x) = \frac{u(x+h) - 2u(x) + u(x-h))}{h^2} + O(h^2) = \frac{u(x+h) - 2u(x) + u(x-h))}{h^2} + O(h^2)$$

Hvorfor?

$$u_x = 3x^2 + e^{-x}, \quad u(0) = 3$$

$$u_x(x_1) = \frac{u(x_0+h) - u(x_0)}{h} + O(h) = 3x_0^2 + e^{-x_0}$$

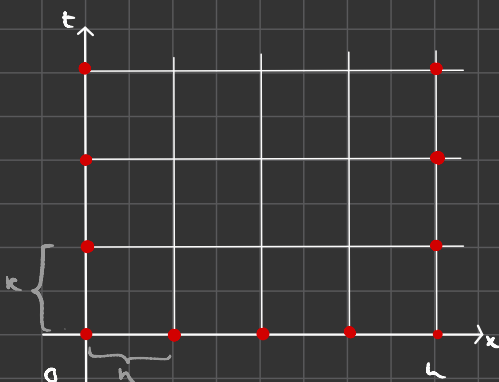
$$u(x_1) = u(x_0) + (3x_0^2 + e^{-x_0})h$$

$$u(x_2) = u(x_1) + (3x_1^2 + e^{-x_1})h$$

For i in $(x=x_0, x=L)$

$$u(x_i) = \dots$$

FDM 1D



$$u(x_i, 0) = f(x_i)$$

$$u(0, t) = c = u(L, t)$$

• kjente pkt

$$u_t = u_x, \quad u(x, 0) = f(x)$$

$$\frac{u(x, t+k) - u(x, t)}{k} = \frac{u(x+h, t) - u(x, t)}{h}$$

$$u(x, t+k) = u(x, t) + \frac{k}{h} (u(x+h, t) - u(x, t))$$

$$= (1 - \frac{k}{h}) u(x, t) + \frac{k}{h} u(x+h, t)$$

$$u(1, 1) = (1 - \frac{k}{h}) u(1, 0) + \frac{k}{h} u(2, 0) \quad \text{iterasjon}$$

$$\begin{array}{c} u(x, t+k) \\ u(x-h, t) \quad u(x, t) \quad u(x+h, t) \\ u(x, t-k) \end{array}$$

EKS:

FDM på $u_t = c^2 u_{xx}$ med forward i tid og central i rom

$$u_t = \frac{u(x, t+k) - u(x, t)}{k}$$

$$u_{xx} = \frac{u(x+h, t) - 2u(x, t) + u(x-h, t)}{h^2}$$

$$u(x, t+k) = u(x, t) + \frac{k}{h^2} (u(x+h, t) - 2u(x, t) + u(x-h, t))$$

Forward Euler på varmeledningen, videre føring av eks over

$$u_t = u_{xx} \quad \begin{array}{l} \swarrow \text{tidssteg} \quad \searrow \text{romsteg} \end{array}$$

$$\frac{u(x, t+k) - u(x, t)}{k} = \frac{u(x+h, t) - 2u(x, t) + u(x-h, t)}{h^2}$$

Endrer notasjon til U_m^n , n = tid, m = rom

$$\frac{U_m^{n+1} - U_m^n}{k} = \frac{U_{m+1}^n - 2U_m^n + U_{m-1}^n}{h^2}$$

Vi ønsker å starte i tiden $t=0$ med kjente initial- og gr.bet., $u(x, 0)$, $u(0, t)$ og $u(L, t)$. Fra

dette må vi la tiden begynne å gå. Vi lurer på hvordan likningen ser ut i neste tidssteg og løser for U_m^{n+1}

$$U_m^{n+1} = \frac{k}{h^2} (U_{m+1}^n - 2U_m^n + U_{m-1}^n) + U_m^n = (1 - \frac{2k}{h^2}) U_m^n + \frac{k}{h^2} (U_{m+1}^n + U_{m-1}^n) \quad \rightsquigarrow \text{iterasjon}$$

I iterasjonen har vi $n \in \{0, \dots, T\}$ som avgjør hvor lenge vi itererer og $m \in \{0, \dots, L\}$ som

betegner lengden på int. Dette kan man sette opp på matriseform

$$\begin{bmatrix} U_1^{n+1} \\ U_2^{n+1} \\ U_3^{n+1} \\ \vdots \\ U_L^{n+1} \end{bmatrix} = \begin{bmatrix} (1 - \frac{2k}{h^2}) & \frac{k}{h^2} & 0 & 0 & \dots & \dots & \dots & 0 \\ \frac{k}{h^2} & (1 - \frac{2k}{h^2}) & \frac{k}{h^2} & \frac{k}{h^2} & 0 & \dots & \dots & 0 \\ 0 & \frac{k}{h^2} & (1 - \frac{2k}{h^2}) & (1 - \frac{2k}{h^2}) & \frac{k}{h^2} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \dots & \dots & \dots & 0 & \frac{k}{h^2} & (1 - \frac{2k}{h^2}) \end{bmatrix} \begin{bmatrix} U_1^n \\ U_2^n \\ U_3^n \\ \vdots \\ U_L^n \end{bmatrix}$$

FDM for bølgelikningen

$$u_{tt} = u_{xx}$$

Endring i
stigningstall

$$u(x+h) = \sum_{n=0}^{\infty} \frac{u^{(n)}(x)}{n!} h^n \Rightarrow u(x+h) = u(x) + u'(x)h + \frac{u''(x)h^2}{2} \Rightarrow u''(x) =$$

$$u(x-h) = \sum_{n=0}^{\infty} \frac{u^{(n)}(x)}{n!} (-h)^n \Rightarrow u(x-h) = u(x) - u'(x)h + \frac{u''(x)h^2}{2}$$

Endringen av
endring i st. tall

$$u(x+h) + u(x-h) = (u(x) + \cancel{u'(x)h} + \frac{u''(x)h^2}{2}) + (u(x) - \cancel{u'(x)h} + \frac{u''(x)h^2}{2})$$

$$u''(x) = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

$$u_{tt}(x,t) = \frac{u(x,t+k) - 2u(x,t) + u(x,t-k)}{k^2} + O(k^2)$$

$$u_{xx}(x,t) = \frac{u(x+h,t) - 2u(x,t) + u(x-h,t)}{h^2} + O(h^2)$$

den vi ønsker å løse for

$$\frac{u(x,t+k) - 2u(x,t) + u(x,t-k)}{k^2} = \frac{u(x+h,t) - 2u(x,t) + u(x-h,t)}{h^2}$$

$$U_m^{n+1} = 2(1 - \frac{k^2}{h^2}) U_m^n + \frac{k^2}{h^2} (U_{m+1}^n + U_{m-1}^n) - U_m^{n-1}$$

Dersom $n=0$ får vi U_m^{-1} som siste ledd

der av store 0, vet

ikke om feilen er positiv

sentraldiff

$$u(t+k) - u(t-k) = (u(t) + u'(t)k + \frac{u''(t)k^2}{2}) - (u(t) - u'(t)k + \frac{u''(t)k^2}{2}) + O(k^3)$$

$$= 2k u'(t) + O(k^3)$$

$$u'(t) = \frac{u(t+k) - u(t-k)}{2k} + O(k^2)$$

$$u_t(x,0) = \frac{U_m^1 - U_m^{-1}}{2k} + O(k^2) \Rightarrow U_m^{-1} = U_m^1 - 2k u_t'(x,0)$$

$$U_m^{n+1} = 2(1 - \frac{k^2}{h^2}) U_m^n + \frac{k^2}{h^2} (U_{m+1}^n + U_{m-1}^n) - U_m^{n-1} + 2k g(x_m)$$

Gjelder kun første rad